

Suggested FYP topics for 2026-27

by

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Project code :	YYC1
Project title :	An EEG-Based BCI Communication System
No. of students :	2
Description of project :	This project aims to develop an EEG-based Brain-Computer Interface (BCI) communication system that enables users to convey messages using brain signals. This system is particularly useful for individuals with severe motor impairments, such as those with locked-in syndrome (LIS) or neuromuscular disorders, who cannot communicate through traditional means. The system will process electroencephalography (EEG) signals to detect user intent and translate brain activity into text or control commands. Transformer or suitable machine learning models will be trained to classify EEG patterns associated with different cognitive or motor imagery tasks. The project will also explore real-time signal processing to enhance system responsiveness and accuracy. Key Features: • EEG-Based Signal Acquisition – Captures brain activity using a non-invasive EEG headset. • Signal Processing & Feature Extraction – Filters and analyzes EEG data to identify relevant brain patterns. • Machine Learning for Classification – Uses deep learning models to map EEG signals to communication commands. • Real-Time Communication Interface – Displays predicted text or symbols for user interaction. Example Applications: • Text-Based Communication System – Converts EEG signals into words or phrases for non-verbal users. • Assistive Control Interface – Enables users to control external devices (e.g., smart home appliances) through brain signals. • Neurofeedback Training – Provides real-time feedback to help users improve BCI control accuracy. By the end of the project, students will develop a working prototype of an EEG-based BCI communication system, demonstrating its feasibility for assistive communication and human-computer interaction.
Industrial collaboration :	Yes (LAURRY AI).

Project code :	YYC2
Project title :	An EEG-Based BCI Application
No. of students :	2
Description of project :	This project aims to develop an EEG-based Brain-Computer Interface (BCI) application designed by your own. The target users are not limited to those with disabilities. Similar to YYC1, your system will process electroencephalography (EEG) signals to detect user brain states. Transformers or suitable machine learning models will be trained to classify EEG patterns associated with different cognitive or motor imagery tasks. The project will also explore real-time signal processing to enhance system responsiveness and accuracy. Key Features: • EEG-Based Signal Acquisition – Captures brain activity using a non-invasive EEG headset. • Signal Processing & Feature Extraction – Filters and analyzes EEG data to identify relevant brain patterns. • Machine Learning for Classification – Uses deep learning models to map EEG signals to communication commands. • Student-Designed BCI Application Interface – Implements a custom application developed by yourselves to translate classified EEG signals into interactive functions, such as control, selection, or communication tasks. Example Applications: • On-Screen Spelling Board – Allows users to select letters, words, or symbols using EEG signals. • Simple Brain-Controlled Game – Lets users control basic game actions through detected EEG patterns. • Smart-Home or Robot Control Demo – Uses EEG commands to control a virtual appliance or a small By the end of the project, students will develop a working prototype of an EEG-based BCI application.
Industrial collaboration :	Yes (LAURRY AI).